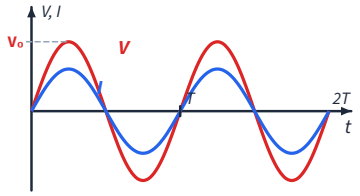
CHAPTER 19 ALTERNATING CURRENT

Key idea: the current and voltage vary sinusoidally and reverse direction every half cycle.

1 BASIC EQUATIONS



Pure resistor: V and I in phase

$$I = I_0 \sin \omega t \qquad V = V_0 \sin \omega t$$

I_0, V_0 peak values

$\omega = 2\pi f$ (rad s^{-1})

$T = \frac{1}{f}$ (s)

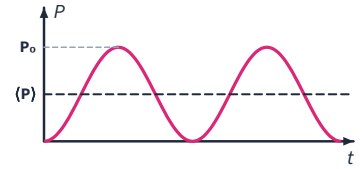
2 R.M.S. (EFFECTIVE) VALUE

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \qquad V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$

- The steady d.c. that dissipates heat in a resistor **at the same rate** as the a.c.
- Mean value over one cycle $= 0$
- Mains 240 V is r.m.s.: $V_0 = 339$ V

3 POWER IN A.C.

$$\langle P \rangle = I_{\text{rms}} V_{\text{rms}} = I_{\text{rms}}^2 R = \tfrac{1}{2} I_0 V_0$$



- Peak power $P_0 = I_0^2 R = 2\langle P \rangle$
- Instantaneous P varies at **twice** the supply frequency, never negative

Pure L and pure C: mean power $= 0$. Energy is stored, then returned to the source.

4 COMPONENTS

RESISTOR R

- V and I **in phase**, $\phi = 0$
- R independent of f
- Dissipates energy as heat

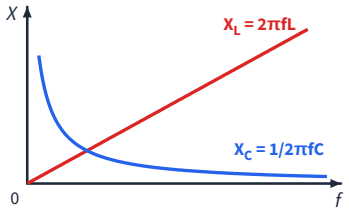
INDUCTOR L

- $X_L = \omega L = 2\pi f L$
- V **leads** I by 90°
- d.c.: $X_L = 0$

CAPACITOR C

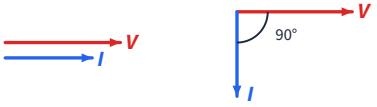
- $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
- I **leads** V by 90°
- Blocks d.c.: $X_C \to \infty$

5 REACTANCE AND FREQUENCY

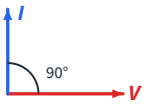


- $X_L \propto f$: straight line through origin, gradient $2\pi L$
- $X_C \propto \frac{1}{f}$: falls towards zero, never touches the axis
- f doubled: X_L doubles, X_C halves

6 PHASE (PHASORS)



R: in phase L: I lags V

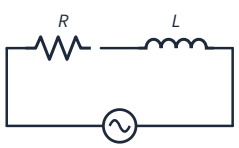


C: I leads V

Series R–L or R–C: phase angle $0 < \phi < 90^\circ$

ELI the ICE man: in L, e.m.f. leads I; in C, I leads e.m.f.

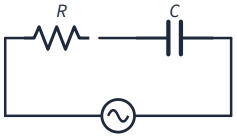
7 R-L AND R-C SERIES CIRCUITS



$$Z = \sqrt{R^2 + X_L^2}$$

$$\tan \phi = \frac{X_L}{R}$$

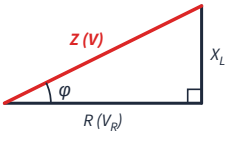
R–L: current **lags** voltage by ϕ



$$Z = \sqrt{R^2 + X_C^2}$$

$$\tan \phi = \frac{X_C}{R}$$

R–C: current **leads** voltage by ϕ



$$V^2 = V_R^2 + V_L^2$$

$$\cos \phi = \frac{R}{Z}$$

$V_{\text{rms}} = I_{\text{rms}} Z$ and mean power $\langle P \rangle = I_{\text{rms}} V_{\text{rms}} \cos \phi = I_{\text{rms}}^2 R$ (only R dissipates power).

8 UNIT CHECK

Quantity	Formula	Unit
Period	$T = 1/f$	s
Angular frequency	$\omega = 2\pi f$	rad s^{-1}
R.m.s. current	$I_{\text{rms}} = I_0/\sqrt{2}$	A
Inductive reactance	$X_L = \omega L$	Ω
Capacitive reactance	$X_C = 1/\omega C$	Ω
Impedance	$Z = \sqrt{R^2 + X^2}$	Ω
Power factor	$\cos \phi = R/Z$	no unit
Mean power	$I_{\text{rms}}^2 R$	W

9 KEY POINTS TO MEMORISE

1. R.m.s. value = peak value $\div \sqrt{2}$ (sine wave only).
2. Only the resistor dissipates mean power.
3. Inductor stores energy in its **magnetic** field; capacitor in its **electric** field.
4. Voltages across R and X are 90° apart: add as **phasors**, not arithmetically.

10 COMMON SECTION A TRAPS

- Reading ω (314 rad s^{-1}) as the frequency: $f = \omega/2\pi = 50$ Hz.
- Using V_0 in $P = V^2/R$ gives the **peak** power, twice the mean.
- Forgetting μF → ×10^{-6} F or mH → ×10^{-3} H.
- Stating that current leads voltage in an inductor (that is the capacitor).

EXAM TIPS

- ✓ Convert peak values to r.m.s. before finding power.
- ✓ Find X_L or X_C first, then Z, then I.
- ✓ Draw the phasor (impedance) triangle: R across, X up, Z hypotenuse.
- ✓ State the phase relation in words: "current leads/lags voltage by …".